

Send the Link: Launch, Demo & Story

Final Capstone Synthesis — Public Launch, Demo Video Proof, and 10–Week Engineering Narrative

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Track: General AI Fluency (Week 10 Capstone)

Live Launch URL: abdulsamiuthwal-portfolio.vercel.app

Live Demo Video: youtu.be/-uCqi_JDjfY

1. Executive Summary & Capstone Vision

A portfolio is not a static PDF or classroom assignment—it is a live, verifiable career platform that proves what you can build, deploy, and defend. This final Capstone deliverable synthesises the three core pillars of the FlyRank General AI Fluency Track: Launch, Demo, and Story.

2. The Three Core Capstone Pillars

PILLAR 1: THE LAUNCH (PRODUCTION INFRASTRUCTURE)

URL: <https://abdulsamiuthwal-portfolio.vercel.app/>

Public deployment over HTTPS via Vercel Edge Network, fortified with Vercel Web Analytics, OpenGraph social cards, SVG favicon, and the embedded FlyRank Verified Graduate Badge.

PILLAR 2: THE DEMO (VIDEO PROOF & AGENT ARCHITECTURE)

Video Link: https://youtu.be/-uCqi_JDjfY (Unlisted YouTube)

Live screen recording demonstrating ResearchScout agent fetching arXiv REST API live, parsing Atom XML, synthesising engineering briefs, and saving to disk in <900ms. Includes verbal explanation of Design Decision (XML vs PDF speedup) and Known Limitation (template synthesis).

PILLAR 3: THE STORY (10–WEEK ENGINEERING JOURNEY)

From heuristic CTR opportunity baselines (F1: 0.480) to Champion Gradient Boosting models (F1: 0.783, ROC-AUC: 0.983), autonomous literature agents, serverless Web3Forms lead capture pipelines, PageSpeed 98/100 hardening, and verified graduate launch.

3. Engineering Milestone Progress Matrix

Phase / Milestone	Core Artifact / Technology	Verified Result / Status
Weeks 1–4: Signal Framing	CTR opportunity gap formulation across 600 pages	Data contract & schemas established
Weeks 5–6: ML Modeling	Gradient Boosting Champion vs 4 Baselines	GroupKFold Validated F1: 0.783 (ROC-AUC: 0.983)
Weeks 7–8: Agent & Lead System	ResearchScout Agent + Web3Forms API	Sub-2s arXiv briefs + live lead email delivery
Weeks 9–10: Launch & Capstone	SEO metadata, Vercel Analytics, FlyRank Badge	PageSpeed: 98/100 , Live HTTPS deployment